

AEM-X100 — Sensor Troubleshooting Guide

Equipment: AEM-X100 Wafer Handling System

Document ID: SYN-X100-SENSOR_TROUBLESHOOTING **Revision:** A **Status:** Synthetic Training Data

IMPORTANT: This document is entirely fictional and created for the AAS Practice Module prototype. It is not an AEM document, does not represent an actual AEM product or maintenance procedure, and must not be used for real equipment servicing.

Purpose: Provide synthetic troubleshooting evidence for wafer-presence and position-sensor faults.

1. Scope

- This fictional guide covers WPS-A, WPS-B and the transfer-position sensor family used by the AEM-X100 prototype.
- The guide does not authorize physical repair or electrical work. Safety Procedure and Escalation Procedure take precedence.

2. Wafer Presence Sensor — WPS-A

- Symptom: SCU reports that a wafer is not detected at the pickup position.
- Check 1 — Confirm the system state and displayed alarm code.
- Check 2 — Confirm the sensor status changes when the documented test condition is applied.
- Check 3 — Check for obvious obstruction or contamination using the approved visual inspection method.
- Check 4 — Confirm sensor alignment indicator is within the fictional green range.
- Check 5 — If the sensor remains inactive, stop repeated transfer attempts and escalate according to the documented escalation path.

3. Position Sensor Fault

- Symptom: WTM reaches a position but SCU does not confirm the expected position.
- Check the displayed error code.
- Review the relevant position-sensor entry in Error Code Reference.
- Verify whether the fault is persistent or occurred during a single transfer attempt.
- Do not bypass an interlock or modify motion limits to force a transfer.

4. Diagnostic Decision Table

Observed condition	Next documented action
WPS status inactive	Follow WPS troubleshooting checks
Sensor active but wafer absent	Verify documented wafer/loading condition
Position not confirmed	Search position-sensor fault and escalate if persistent
Safety interlock active	Stop troubleshooting and follow Safety Procedure
No matching evidence	Use Escalation Procedure

5. Cross-References

- Error codes: Error Code Reference.
- Protected access or interlock condition: Safety Procedure.
- Unresolved or repeated fault: Escalation Procedure.

End of synthetic document.